QP Code: _____

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR PRACTICAL EXAMINATION MODEL QUESTION PAPER

PHYSICS OF ADVANCED MATERIALS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 35**

Record: 5 Marks

1. From the given experimental data, perform the thermal analysis of the material. Plot the appropriate graph, carry out the necessary calculations, and determine the required thermal parameter(s). [A]/[CO6]

2. Study the variation of resistance with temperature for a carbon film resistor, metal wire, semiconductor and thermistor. Plot suitable graphs and compare their temperature dependence. [A]/[CO6]

3. Using the given absorption-spectrum data, determine the band gap energy of the material and identify whether the optical transition is direct or indirect using the Tauc relation. Show all relevant calculations and graphical analysis. [A]/[CO6]

4. Determine the absorption coefficient of the given solution and study its dependence on path length and concentration using the supplied experimental data. [A]/[CO6]

5. Determine the specific rotation of the given optically active substance from the experimental observations. Include the relevant formula, observations, calculation and final result with proper units. [A]/[CO6]

6. Determine the frequency dependence of the dielectric constant of the given material. Tabulate the observations, perform the required calculations and represent the variation graphically. [A]/[CO6]

7. Determine the temperature dependence of the dielectric constant of the given material. Record the observations, calculate the dielectric constant and plot the variation with temperature. [A]/[CO6]

8. Using Lee's disc method, determine the thermal conductivity of the given bad conductor. Include all measurements, formulae, calculations and the final result with proper units. [A]/[CO6]

9. Determine the dielectric constant of the given thin sheet using the prescribed experimental method. Show the observations, calculations and final result. [A]/[CO6]

10. Determine the refractive index of the given glass using the liquid-lens method or spectrometer method. Include the necessary observations, calculations and final result. [A]/[CO6]

11. Using the prescribed bending/elongation method, determine the Young's modulus of the given polymer composite (wood). Include observations, calculations and the final result with proper units. [A]/[CO6]

12. Draw the B–H hysteresis curve of the given ferromagnetic material and determine its retentivity and coercivity from the curve. Clearly indicate the relevant quantities on the graph. [A]/[CO6]

13. Perform the Hall-effect experiment on the given semiconductor and determine the Hall coefficient, carrier concentration and mobility. Include observations, calculations and final results with proper units. [A]/[CO6]

14. Using the four-probe method, determine the resistivity of the given semiconductor and hence determine its energy band gap from the temperature dependence of resistivity. Show all observations, graphs, calculations and final results. [A]/[CO6]

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) V SEMESTER PRACTICAL EXAMINATION

DSE: MG5DSEPHY307 - OP AMP AND LINEAR INTEGRATED CIRCUITS

Duration: 2 hrs **Maximum Marks: 35**

Record: 5 Marks

Attempt the question marked 'X'. Change of question is not allowed (30 marks)

1. Design and construct an op-amp based square-wave generator. Generate a square wave of the specified frequency and verify the output waveform using CRO/DSO. [A, An, S, C][5]
2. Design and construct a first-order active low-pass filter using an op-amp. Analyse the variation of gain with frequency by plotting the frequency response graph. [A, An, S, C][5]
3. Design and construct a first-order active high-pass filter using an op-amp. Analyse the variation of gain with frequency by plotting the frequency response graph. [A, An, S, C][5]
4. Construct an op-amp based Pulse Width Modulation (PWM) circuit. Generate a PWM signal and study the variation of pulse width with the input control voltage. [A, An, S, C][5]
5. Construct and test an op-amp based A/D converter. Apply suitable analog input voltages and verify the corresponding digital output. [A, An, S, C][5]
6. Design and construct an op-amp summing amplifier. Apply two or more inputs and verify the relationship between the input voltages and the output voltage. [A, An, S, C][5]
7. Design and construct inverting, non-inverting and buffer amplifiers using op-amp. Obtain the voltage gain for each configuration using AC input voltages. [A, An, S, C][5]
8. Design and construct an op-amp differential amplifier. Apply two input signals and verify the differential amplification by measuring the output voltage. [A, An, S, C][5]
9. Construct an op-amp integrator circuit. Apply a suitable input waveform and observe the integrated output waveform. Verify the theoretical relationship between input and output. [A, An, S, C][5]
10. Construct an op-amp differentiator circuit. Apply a suitable input waveform and observe the differentiated output waveform. Verify the theoretical relationship between input and output. [A, An, S, C][5]
11. Construct an op-amp current-to-voltage (I–V) converter. Verify the current-to-voltage conversion relationship by applying different input currents and measuring the corresponding output voltages. [A, An, S, C][5]
12. Construct and study an op-amp voltage follower. Apply different input voltages and verify that the output voltage follows the input voltage with unity voltage gain. [A, An, S, C][5]